Week 5 (16.02.2026)

① A : J ≻ I ≻ K
B : I ≻ J ≻ K
C : I ≻ J ≻ K

I : A ≻ B ≻ C
J : B ≻ A ≻ C
K : A ≻ B ≻ C

Let's find a stable matching with the Gale-Shapley algorithm with men proposing.

1. round

I and K propose to A;
A tentatively accepts I.

J proposes to B;
B tentatively accepts J.

2 round

K proposes to B;
B rejects K.

3 round

K proposes to C;
C accepts K.

Final matching:

$\boxed{\{A-I, B-J, C-K\}}$

Let's find a stable matching with women proposing:

round 1

A proposes to J,
B and C propose to I;

J accepts A,
I accepts B.

round 2

C proposes to J;
J rejects C.

round 3

C proposes to K;
K accepts C.

Final matching:

$\boxed{\{B-I, A-J, C-K\}}$

② slides

③ Is Gale-Shapley strategyproof?

If men are the default proposing side of the GS algorithm, then:

Theorem: on every instance of the stable matching problem, no man can benefit from misreporting their preferences under GS – i.e., GS is strategyproof for men.

Proof: one can derive the theorem directly from the fact that GS outputs the men-optimal matching.

Strategyproofness for all?

Claim: under Gale-Shapley, women can benefit from misreporting their preferences

Gale-Shapley: $\boxed{\{A-I, B-J, C-K\}}$

Suppose A misreports as $J \succ K \succ I$. Then:

I proposes to A: $\{A-I\}$
J proposes to B: $\{A-I, B-J\}$
K proposes to A; she rejects I and accepts K: $\{A-K, B-J\}$
I proposes to B; she rejects J and accepts I: $\{A-K, B-I\}$

J proposes to A; she rejects K and accepts J: {A-J, B-I}

K proposes to C: {A-J, B-I, C-K}

<u>Claim</u>: there is no algorithm that always outputs stable matchings such that no agent (M or W) can benefit from misreporting their preferences.

(4) Matching with one-sided preferences

In stable matching, two sides both have preferences over each other. In some settings, only one side has preferences, and the other side is indifferent:

- matching individual students to rooms;

- matching (equally competent) workers to tasks.

House allocation problem:

- a set of n agents N
- a set of n houses H
- each person in N has a ranking over all houses in H
- goal: match agents to houses (1:1)

House allocation: no endowments

Case 1: initially all houses are unassigned

A : $x > y > z > t$
B : $x > t > y > z$
C : $t > z > y > x$
D : $z > x > y > t$

Serial dictatorship:

- agents come one by one
- each agent chooses her most preferred available house
- the agent and the house are removed from the market

Outcome:

A: x > y > z > t
B: t > y > z
C: z > y
D: y

Is this:
- strategyproof?
- core stable?
- fair?
- ensures a good quality allocation?

Serial dictatorship: properties

Claim: serial dictatorship is $\boxed{\text{strategyproof}}$

- each agent gets their most preferred available house

> An allocation is $\boxed{\text{core-stable}}$ if no group of agents can deviate by exchanging houses among themselves so that each deviator prefers the new allocation

Claim: serial dictatorship is core-stable

Proof: suppose there is a deviating coalition S, and let i be the first agent in S. Let h be the house that i gets under SD. When i chose h, agents in $S \setminus \{i\}$ have not made

their selection, so their houses were still available – hence, i prefers h to houses of agents in $S \setminus \{i\}$. $\square$

An allocation A is $\boxed{\text{Pareto-dominated}}$ by allocation B if:

- no agent is worse off in B compared to A
- some agent is strictly better in B compared to A

An allocation is $\boxed{\text{Pareto-optimal}}$ if it is not Pareto-dominated by another allocation

<u>Claim:</u> the output of SD is Pareto-optimal

<u>Proof:</u> Let A be the output of SD, and suppose

B Pareto-dominates A.
Let i be the first agent (in the order of SD) who gets different houses in A and B, and let h be her house in A.
Let T be the set of houses that SD allocates to agents $\{1, \dots, i-1\}$. In B, agents in $\{1, \dots, i-1\}$ also get houses in T. Hence, in B agent i gets a house in $H \setminus T$ — but h is her favourite house in $H \setminus T$!!! $\square$

$\boxed{\text{Serial dictatorship is } \underline{\text{not}} \text{ fair}}$

<u>Fix:</u> draw a random order of agents, let them choose in that order.

House allocation with endowments

Case 2: each agent initially owns a house
- need <u>not</u> rank her house first.

A : x > y > z > t
B : x > t > y > z
C : t > z > y > x
D : z > x > y > t

Can we reduce this to the case without endowments?

Attempt: evict all agents, run SD
- may fail individual rationality:
  - an agent may end up with a house she likes less than her original house
  - she will then prefer not to participate

Attempt: allow mutually beneficial pairwise exchanges

- may fail Pareto optimality, core stability

Attempt: allow mutually beneficial group exchanges

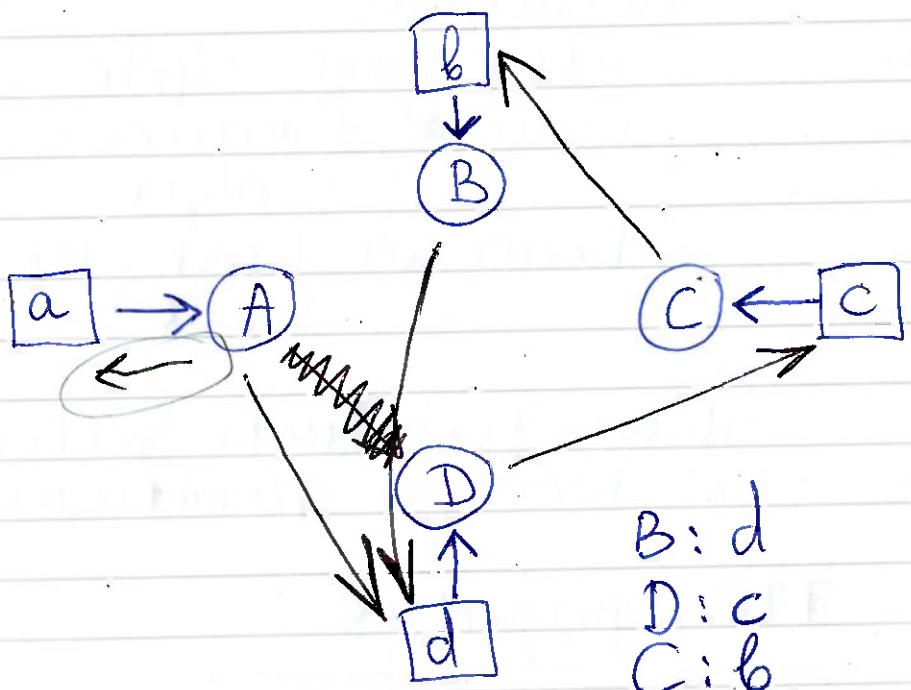
- seems expensive to implement

Algorithm: Top Trading Cycle (TTC)

- Each person points to most preferred house
- Each house points to its owner
- This creates a directed graph, with $2n$ vertices and $2n$ edges, so at least one cycle

- Remove all cycles, assigning people to the house they are pointing at
- Repeat using preference lists where the assigned houses have been deleted

A : d > a > c > b
B : d > b > a > c
C : b > c > a > d
D : c > d > a > b



B : d
D : c
C : b
A : a

Does TTC algorithm always terminate?

Yes. Initially, there are $2n$ vertices and $2n$ edges, hence at least one cycle (possibly of length 2)

Each iteration removes a cycle (at least 2 vertices)

Induction:
- after each cycle removal, # vertices = # edges
- hence, at least one cycle

Hence, terminate after at most n iterations.

TTC: properties

Theorem: TTC is <u>individually rational</u> – no agents receives a house that they like less than their original house

Proof: Consider an agent i that originally has house h. Suppose TTC allocates h' to i and i prefers h to h'.

Consider the round in which i receives h'. In this round, h is still present (we only remove a house together with its owner). But then, i cannot be pointing to h'. $\square$

Theorem: TTC outputs a <u>Pareto-optimal</u> allocation.

Proof: Let X be the allocation output by TTC. Suppose Y Pareto-dominates X.

Let N^* be the set of agents who improve in Y.
Let k be the first round in which some agent i in N^* gets a house under TTC.
Let H^* be the set of houses allocated before round k.
The houses in H^* are allocated identically in X and Y. Hence, in Y agent i gets a house in $H \setminus H^*$ – but in X, i gets her most preferred house in $H \setminus H^*$! $\square$

An allocation X is in the core if there is no set of agents $N' \subseteq N$ that can trade the houses they originally own so that each of them prefers the resulting allocation to X.

<u>Theorem</u>: TTC outputs an allocation in the <u>core</u>.

<u>Proof</u>: let X be the output of TTC.

Suppose there is a coalition N' that can beneficially deviate.

Let k be the first round in which some agent $i \in N'$ gets a house under TTC, and let h be the house that i gets in TTC.

Let h' be the house that i gets when deviating with N'. All agents in N' are present in round k. Hence, so are their houses – in particular, h'. Since i points to h, she prefers h to h'. $\square$

<u>Theorem</u>: TTC is <u>strategyproof</u>.

<u>Proof sketch</u>: Fix an agent i.

We can think of i as choosing which house to point to in each round.

If agent i is truthful, she points to her favourite house.

Why would someone point to a house which is <u>not</u> their favourite?
- Because they can then get that house in this round, and they worry that later it may disappear

Let $H(K)$ be agent i's "choice set" in round K

<u>Lemma</u>: $H(K) \subseteq H(K+1)$
- every vertex has outdegree 1
- if an agent j has path to i in round K, this remains the

case in round k+1 (until i is allocated) □

House allocation with endowments = kidney exchange

Roth, Sonmez and Unver (2004)

- The problems are (essentially) equivalent
- TTC can be used to efficiently assign kidneys

In 2004, RSU and doctors in Boston established first clearinghouse for New England.